

Lyman- α Blobs (LABs) are giant Ly α nebulae extending up to 100–300 kpc at high redshift. They trace the complex interplay between galaxies, their circumgalactic medium (CGM), and the surrounding intergalactic medium (IGM) in the cosmic web at epochs of peak cosmic star formation and AGN activity ($z \sim 2\text{--}4$), and host the baryonic cycle—accretion, outflow, and gas recycling—in action (Cantalupo et al. 2014, Ouchi et al. 2020). The CGM and IGM gas around these overdensities is the primary fuel reservoir for galaxies and the repository of feedback products. Despite their discovery over two decades ago, the dominant physical mechanisms powering Ly α emission and governing their gas kinematics remain to be explored.

Lyman- α emission from LABs and Ly α emitters (LAEs) is predominantly powered by (i) cooling radiation from cold gas accretion, (ii) photoionization by embedded sources such as star-forming galaxies or AGN, or (iii) shock heating associated with galactic-scale outflows. Each of these mechanisms imprints distinct signatures on the Ly α line profile and spatially resolved kinematics. Along line of sight, the integrated Ly α profiles are shaped by gas motion into single-peaked, double-peaked (with red or blue dominance), and asymmetric shapes, each linked to a dominant gas motion (see Fig. 1). Blue-dominant double-peaked profiles are strong indicators of inflowing gas, as photons scatter in a medium accelerating toward the observer, suppressing the red peak (Mukherjee et al. 2026; Bolda et al. 2025). Conversely, red-dominant profiles are classic signatures of expanding shells or outflows (Herenz et al. 2025; Pahl et al. 2024). The separation between the two peaks ($\Delta\nu$) scales with the neutral hydrogen column density, providing a means to estimate gas content (Li & Zheng 2025; Blaizot et al. 2023). Resonant scattering of Ly α photons further complicates the interpretation.

Most existing studies of LABs or LAEs have focused on individual objects or have used low-to-moderate spectral resolution ($R \sim 2000\text{--}3000$). At these resolutions, the double-peaked structure is often blended in individual spaxels, preventing robust classification of inflow versus outflow regions and accurate measurement of peak separation (Vitte et al. 2025; Tornotti et al. 2025). Moreover, Ly α escape is intrinsically anisotropic: the observed profile depends on the viewing angle relative to the local gas velocity gradient (Almada Monter et al. 2026). Therefore, only high-resolution ($R \gtrsim 6000$) spectroscopy can decode the true gas kinematics across a nebula.

Yet, a critical gap remains: the field lacks a systematic, high-spectral-resolution kinematic survey of a statistically significant sample of LABs and their cosmic environments. Such a survey would provide the first robust census of how inflows and outflows are distributed within these massive halos and how they relate to the properties of the central galaxies and the large-scale environment.

This proposal is designed to directly address the gaps outlined above. **We propose to obtain deep spectroscopic observations of a carefully selected sample of about 50 luminous LABs and 1,600 LAEs using the multi-object spectrograph M2FS mounted on the Magellan Clay telescope over 2 nights**, enabling detailed measurements of Ly α line profiles, including velocity offsets, line asymmetries, and multiple components, with sufficient sensitivity to probe gas motion along line of sight. The samples of LABs and LAEs are selected from our ongoing project, in which 861 LAEs at $z=3.12$ and 1101 LAEs at $z=3.20$ have been identified in the J2259 and J0022 field, respectively, through deep narrow-band imaging with CFHT/MegaCam. Of them, 33 and 28 are LABs. These LAEs map the large-scale density fields, revealing that these LABs mostly reside in relatively more dense environment.

The proposed spectroscopic observations using **2 dark nights with Clay/M2FS** will allow us to achieve the following key science goals:

(1) Characterize the multi-phase kinematics of the CGM/IGM around LABs and LAEs: We will measure Ly α velocity centroid, velocity dispersion, Ly α peak ratio (blue/red), and

double-peak separation $\Delta\nu$ for ~ 50 LABs and $\sim 1,600$ LAEs. These measurements will shed light onto the complex interplay between inflowing, outflowing, and quiescent gas on scales of a few kpc, providing the first high-resolution kinematic atlas of LABs and LAEs along line of sight, thereby constraining the physical origin of $\text{Ly}\alpha$ emission and the role of LABs and LAEs in galaxy formation and baryon cycling (Blaizot et al. 2023). Constraining the kinematics of gas in these systems will shed light on how galaxies acquire their gas, how they regulate star formation. These processes are fundamental to the evolution of galaxies across cosmic time (Kereš et al. 2005).

Moreover, such large samples of LABs and LAEs will enable a quantitative comparison with state-of-the-art cosmological simulations (Anglés-Alcázar et al. 2017), and quantify the relative importance of cold gas accretion, galactic winds, and radiative transfer effects imprinted in the $\text{Ly}\alpha$ profiles. We will directly test the predictions of galaxy formation simulations regarding the relative importance of cold-stream accretion versus feedback-driven outflows in massive halos at $z \sim 2-3$ (Dekel et al. 2009).

(2) Perform CGM tomography using the $\text{Ly}\alpha$ double-peak separation: Using double-peak separation $\Delta\nu$ as a proxy for neutral hydrogen column density (Li & Zheng 2025), we will derive averaged $N(\text{HI})$ for sample LABs. This novel approach, calibrated by radiative transfer simulations, will enable a direct comparison with the gas content predicted by cosmological simulations and provide new constraints on the distribution of cool gas in the CGM.

(3) Probe the connection between LABs and the large-scale environment: Many LABs are found in overdense regions, such as protoclusters, where gas accretion rates and galaxy interactions are enhanced. The proposed observations will spectroscopically identify numerous LAEs and map the density and dynamic environment around sample LABs. We can test whether LABs trace filamentary inflows from the cosmic web or are instead dominated by local processes within massive halos. The presence of coherent, large-scale velocity structures would support a scenario in which LABs are illuminated by cold streams feeding massive galaxies, while more chaotic kinematics would favor feedback-driven origins (Uzsoy et al. 2025).

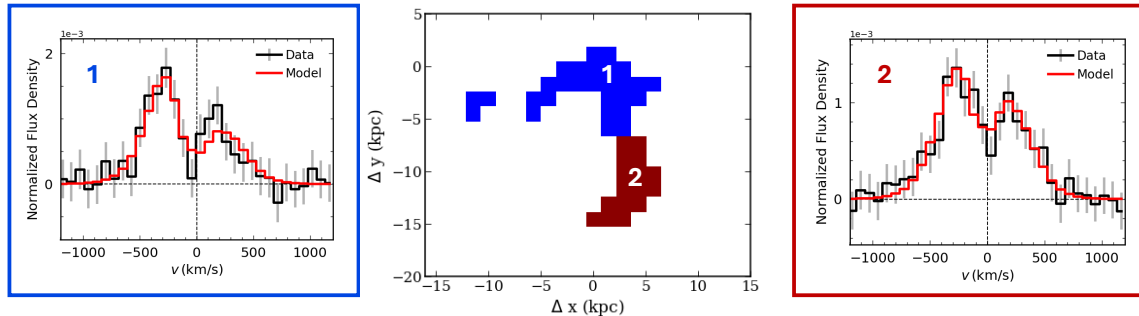


Figure 1: The blue-dominant double-peaked $\text{Ly}\alpha$ line profiles from an extended $\text{Ly}\alpha$ system detected by VLT/MUSE (Mukherjee et al. 2026), suggestive of gas inflows into the target galaxies.

References:

Almada Monter, S., et al., 2026, MNRAS, 547, 330; Anglés-Alcázar, D., et al., 2017, MNRAS, 470, 4698; Blaizot, J., et al., 2023, MNRAS, 523, 3749; Bolda, C., et al., 2025, APJ, 982, 54; Cantalupo S. et al. 2014, Nature, 506, 63; Dekel, A., et al., 2009, Nature, 457, 451; Herenz, E. C., et al., 2025, A&A, 693, A252; Kereš, D., et al., 2005, MNRAS, 363, 2; Li, P. & Zheng, Z., 2025, MNRAS, 542, 2151; Mukherjee T. et al. 2026, A&A, arXiv:2603.11411; Nianias, J., et al., 2025, arXiv e-prints, arXiv:2506.20422; Ouchi, M., et al., 2020, ARA&A, 58, 617; Pahl, A., et al., 2024, ApJ, 974, 212; Uzsoy, A. S. M., et al., 2025, arXiv e-prints, arXiv:2511.17498; Vitte, E., et al., 2025, A&A, 694, A100

This program proposes the use of the Magellan/M2FS to obtain high-resolution spectroscopic observations for 30 luminous LABs. M2FS provides a 30-arcmin diameter field of view and supports 256 science fibers, each with a 1.2'' aperture. This wide field is crucial for capturing the large-scale environmental context of LABs, including satellite galaxies and the cosmic web filaments that may connect them. For the proposed kinematic study, the spectrograph will be configured in its medium-to-high resolution modes ($R \approx 10,000$ – $20,000$). This resolution is necessary to measure velocity offsets with a precision of ~ 20 – 30 km s^{-1} , allowing for the detection of subtle inflow signatures and the measurement of peak separations in low-column-density gas.

Target Selection

For this program, we will set up 8 configurations with M2Fs evenly split in the two fields, J2259 and J0022. In each configuration, the fibers will be deployed to target 6–8 LABs and approximately 200–230 associated LAEs and sky positions per field. The use of 1.2-arcsec fibers is well-matched to the typical sizes of high-redshift galaxies and provides sufficient spatial sampling to resolve the inner and outer kinematics of the extended nebulae.

Exposure Time Estimate

For each pointing, an on-source integration time of 2 hrs are required to measure $\text{Ly}\alpha$ line profiles with integrated fluxes $> 2 \times 10^{-17} \text{ erg s}^{-1} \text{ cm}^{-2} \text{ Hz}^{-1}$.

Instrument Configurations

Redshift Range: $z=3.12$ – 3.20 , where $\text{Ly}\alpha$ is redshifted into the high-efficiency window of the M2FS optics (370–950 nm).

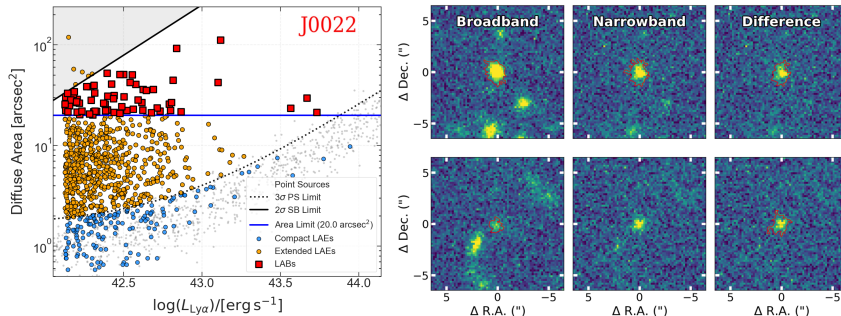


Figure 2: Left: Selection for $\text{Ly}\alpha$ Blobs in the field J0022. Right: Deep narrow [OIII] and broad g images of two $\text{Ly}\alpha$ Blobs in J0022, taken with CFHT/MegaCam (Xu et al. 2026).

CURRENT STATUS OF THE PROJECT AND STUDENT THESIS (Anonymous)

This is a new project.

Abstract

The Lyman- α emission from compact emitters (LAEs) and extended Blobs (LABs; up to ~ 100 – 300 kpc) at $z=2$ – 4 probes the baryon cycle linking galaxies, the CGM, and the IGM. The physical mechanisms powering the Ly α emission and governing gas kinematics remain unclear, largely due to the lack of high-resolution spectroscopic surveys. We propose a systematic study of ~ 50 LABs and 1,600 LAEs using the M2FS spectrograph on the Magellan Clay telescope over 2 nights. With spectral resolution $R=6000$, we will resolve the integrated Ly α line profiles along line of sight, measuring velocity structure, asymmetries, and double-peak separations. The proposed observations will allow us to (1) disentangle CGM kinematics into inflow and outflow signatures, (2) constrain neutral hydrogen column densities, and (3) link gas kinematics to large-scale density and dynamic environment. This program will provide the first high-resolution kinematic census of LABs in relation to their dynamic environment, offering key constraints on gas flows and galaxy formation in the early universe.